



# 2537 - Analysis of Low-Albedo Asteroids Observed through JWST Guaranteed Time

Cycle: 1, Proposal Category: AR

## INVESTIGATORS

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## OBSERVATIONS

### ABSTRACT

We propose to do analysis of the low- albedo asteroid data being collected as part of JWST Guaranteed Time Programs #1244 and #1245: NIRSpec Prism spectra for the Trojan asteroids Hektor and Patroclus and both NIRSpec Prism spectra and MIRI MRS spectra for the main-belt asteroids Ceres, Pallas, and Hygiea and the near-Earth asteroid Phaethon.

The data will be used to address the following broad science questions, as detailed below:

1. What is the range of hydrated mineral composition and amount of hydrated material present on the surfaces of low-albedo asteroids?

As well as a “data handling” question:

2. How can useful spectral information best be extracted from saturated JWST data?

The GTO programs mentioned do not include support for analysis, have no exclusive access period and are open for AR proposals in Cycle 1 (see “JWST Cycle 1 GTO Observations Available for Archival Proposals” in JDox). Support for any analysis of these data will need to come either from this program or from some alternate source that considers it in bounds. In addition, the tasks we propose (extraction of spectra from saturated data and modeling of the thermal emission from the objects to allow separate study of the reflectance and emission spectra) will not be part of a standard JWST reduction pipeline, but will be necessary to allow the proper interpretation of the NIRSpec data and any use of the main belt asteroid MIRI data.

### **OBSERVING DESCRIPTION**

This is a proposal to analyze data to be obtained through GTO programs 1244 and 1245. No observations are requested as a part of this program.